

SEDAR 52 Gulf of Mexico Red Snapper

Life History

- Additional spawning fraction and fecundity collections from all areas of the Gulf have been called for in previous SEDARs (e.g., SEDAR 31). In response, more data have been collected (e.g., SEDAR 52- WP-15). This SEDAR echoes past requests and calls for such data collection to continue to expand in time and space.
- Additional research and analysis is necessary to clarify regional reproductive and demographic differences and, as importantly, trends over time.
- Additional work is needed to improve the SS3 model regarding incorporation of spatial and temporal changes in life history data.

Recreational Discards

- Given the increasing magnitude of removals from the recreational closed season fleets, attempts to improve discard estimates and obtain age composition samples (to help estimate selectivity patterns in the assessment) should be undertaken
- Use of the relative magnitude of estimated charter and private boat discards should be explored to appropriately weight available length composition data from FWC observer survey program within the combined charter-private boat fleet.

Surveys

- Develop SEDAR best practices for recreational CPUE standardization
- Explore removal of CPUE indices from the red snapper assessment
- For the NMFS bottom longline survey conduct paired j and circle hook sets to estimate the hook type effect so as to be able to use the older data from when only j-hooks were used
- Assign current and historical age and length data to habitat type at the finest resolution possible to investigate whether differential catch rates occur among artificial and natural reefs.
- Examine survey design changes during NMFS bottom longline surveys (e.g., the use of Dauphin Island data in the index)
- Examine survey design changes during SEAMAP Groundfish surveys
- Further examine survey design for the combined video index to determine whether it can be used in future red snapper assessments
- Evaluate the use of the SEAMAP vertical line survey in the red snapper assessment as the timeseries expands

Age Composition

- Explore and update the modal analysis methods used to determine the age composition of the SEAMAP groundfish trawl survey length frequency and the shrimp bycatch observer length frequency
- Investigate fitting length composition data directly within the SS3 model as opposed to developing age-length keys and converting length frequency to age composition external to the modeling process

Recruitment

- Explore alternate parametrizations of the stock-recruit function and/or develop informative priors for these parameters
- Incorporate the CMS recruitment index into the base model and investigate ways to use the index to improve recruitment forecasting in the projections

Spatial Modeling

- Explore the potential for developing a fully spatial model of red snapper that can account for differential recruitment and life history patterns across the Gulf of Mexico including differential dynamics on and around artificial versus natural reef habitat

Juvenile Mortality

- Further explore the relationship among shrimp bycatch and juvenile red snapper mortality with emphasis on investigation of incorporating the potential for density-dependent juvenile mortality